ENERGY · ON-PREMISES AI PLATFORM BUILD

Shift Sense

An AI-Powered Shift Intelligence & Digital Logbook Platform for LNG Operations

Turning thousands of free-text operational log entries into instant shift summaries, source-cited answers and early-warning insight — engineered for a fully air-gapped, safety-critical plant, in English and Arabic.

INDUSTRY	SCOPE	ENGAGEMENT
Energy — LNG plant operations (Middle East)	AI summaries, bilingual assistant, actions, analytics & reporting	End-to-end build (design → UAT), fully on-premises

01 — ABOUT THE CLIENT

Where every 12-hour shift ends in a critical handover

The client is a leading LNG producer in the Middle East, operating a multi-train liquefaction plant that runs around the clock on 12-hour shifts. Every shift generates a dense trail of digital logbook entries — activities, critical alarms, safety observations and follow-up actions — held in the plant's electronic logbook and enterprise data platform. A bilingual workforce of operators, supervisors and superintendents depends on that record twice a day, when one crew hands the plant over to the next.

02 — THE CHALLENGE

Four barriers between the logbook and the decision

Slow, manual handovers

Writing and reading shift summaries by hand consumed valuable time at the busiest moment of every shift — and quality depended on the individual.

Buried operational knowledge

Answers lived inside years of free-text log entries. Finding "when did this trip last happen?" meant scrolling history, not asking a question.

Actions lost between shifts

Follow-up actions crossed shift boundaries without clear ownership or tracking, so recurring issues kept resurfacing unnoticed.

Strict data boundaries

An air-gapped perimeter ruled out cloud AI entirely — and the platform had to serve English and Arabic users as equals, with full auditability.

03 — THE SOLUTION

Three AI modules on one governed platform

Soft Suave delivered ShiftSense as a standalone, read-only intelligence layer over the plant's existing logbook and enterprise data platform — an on-premises LLM with retrieval-augmented generation, a near-real-time Spark-to-Iceberg data pipeline, and role-based dashboards for nine personas. It never writes back to source systems.

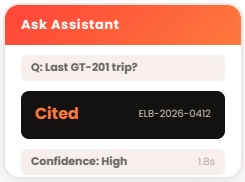
01 Shift Summaries & Handover

Auto-generated at end of shift and on demand — activities, critical alarms, pending actions and safety observations, on the plant's own template, exportable to PDF, Excel and Word.



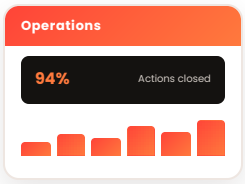
02 Bilingual AI Assistant EN + AR · VOICE

Plain-language questions over the full operational history — typed or spoken — with every answer citing shift date, timestamp and record ID, and a click-through to the original entry.



03 Actions, Trends & Reporting

A full pending-action lifecycle with overdue alerting, recurring-issue detection, KPI and trend dashboards with drill-down, predictive insight, and scheduled management reports.



One on-premises AI brain — summaries, answers and insight, with every response traceable to a dated, timestamped source record.

04 — KEY FEATURES

Twelve capabilities, one platform



AI Shift Summaries (End-of-Shift + On-Demand)



Bilingual AI Assistant (English + Arabic, Text & Voice)



Source-Cited Answers (Date · Timestamp · Record ID)



Pending-Actions Module (Lifecycle & Overdue Alerts)



Analytics, Trends & Predictive Insight



On-Demand & Scheduled Reports (PDF / Excel / Word Export)



In-App & Email Notifications



Role-Based Access & Custom Roles (RBAC)



Dashboard Configuration & Personalization



Near-Real-Time Data Pipeline (~1-Minute Refresh)



Immutable Audit Trail & AI Observability (RAGAS)



Feedback & Continuous Improvement (Human-Gated)

05 — RESULTS & IMPACT

Built to move the numbers that matter on the plant floor

≥95%
AI answer accuracy — English and Arabic measured independently

<3 sec
Assistant response latency on plant-wide queries

~1 min
From new log entry to searchable, AI-ready insight

500+
Concurrent users supported without degradation

Figures represent contracted acceptance targets / by-design outcomes of the platform.

WHY SOFT SUAVE

AI-platform depth, delivered with discipline

The client demanded far more than a chatbot — a production LLM platform inside a zero-egress perimeter, Arabic treated as a first-class language with its own accuracy gate, a near-real-time lakehouse pipeline, and governance aligned to ISO/IEC 42001, NIST AI RMF, OWASP LLM Top 10 and ISO/IEC 27001. Soft Suave delivered the architectural depth and delivery discipline to make it real: every answer source-cited, every interaction immutably audited, and every improvement to the AI human-approved and quality-gated before it ships.

Soft Suave
A KiwiTech Affiliate Company

Soft Suave Technologies Pvt. Ltd.
Chennai, India

softsuave.com
Let's build what's next.